



Moove Education Platform

Technical Design Document (TDD)

Version 3.1

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0.1 API Specification

The platform exposes REST APIs (EdX API v1, OAuth2) with security controls aligned to INSA requirements. Key endpoints are documented below.

0.1.1 Authentication and Authorization

- All API endpoints require authentication via OAuth2 or JWT (CMCSRS 6.17).
- Basic Authentication is disabled in production.
- Access tokens have a limited lifetime and are refreshed using refresh tokens.
- Role-based access control (RBAC) is enforced for all endpoints.

0.1.2 Rate Limiting and Throttling

- Rate limiting is applied to prevent brute-force attacks (fail2ban and Django throttle).
- Login endpoints are limited to 5 attempts per minute per IP.

0.1.3 Input Validation and Output Encoding

- All user input is validated and sanitised to prevent SQL injection, XSS, and command injection.
- Output is properly encoded to prevent XSS.

0.1.4 Key Endpoints

- `/api/user/v1/` – User management (requires OAuth2).
- `/api/courses/v1/` – Course listing and enrollment (requires authentication).
- `/api/gradebook/v1/` – Grade retrieval (requires appropriate role).
- `/api/discussion/v1/` – Forum threads and posts (requires authentication).
- `/api/analytics/v1/` – Usage reports (instructor only).
- `/api/certificates/v1/` – Certificate generation (requires verified enrollment).

For detailed API schemas, see the Swagger documentation (`swagger-docs.json`).

0.2 Database Schema

The database design incorporates security and privacy by design.

0.2.1 Entity-Relationship Diagram

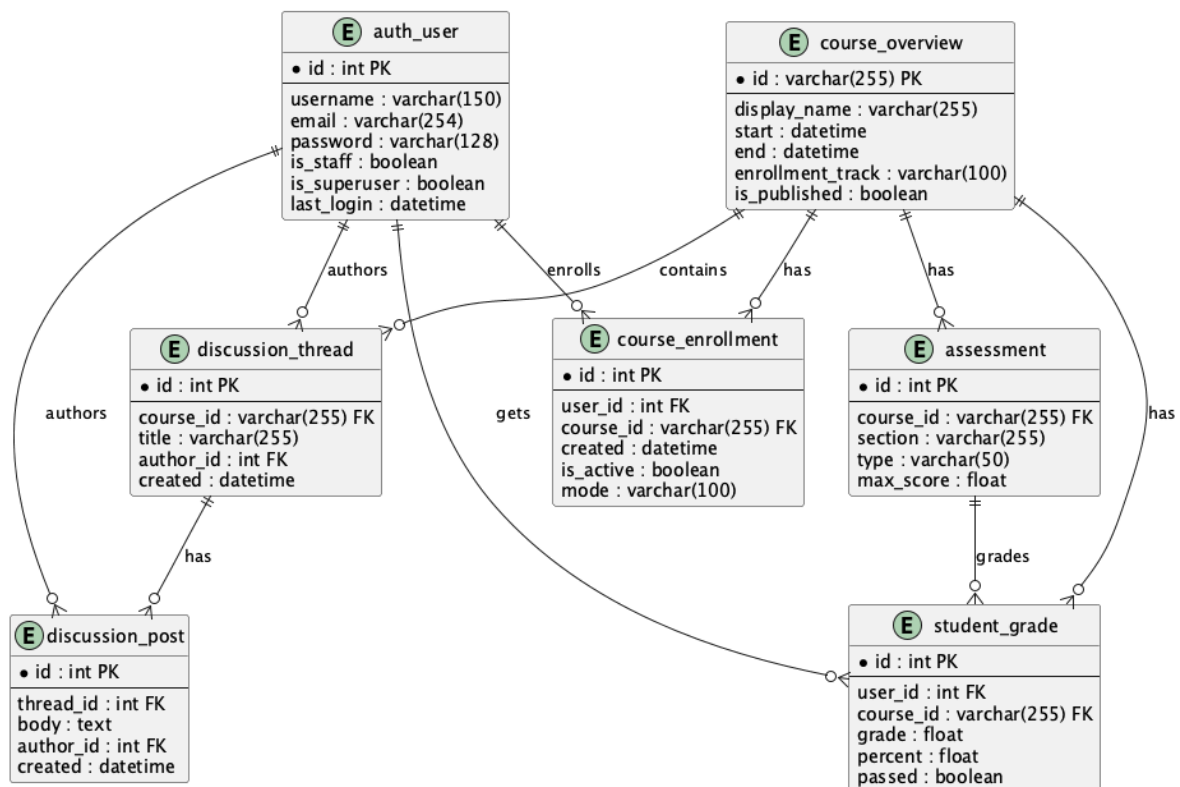


Figure 1: ER Diagram showing primary tables.

0.2.2 Key Tables with Security Considerations

- **auth_user**: `id`, `username`, `email`, `password_hash` (bcrypt), `is_staff`, `is_superuser`.
- **course_overview**: `id`, `display_name`, `start`, `end`, `enrollment_track`.
- **course_enrollment**: `id`, `user_id`, `course_id`, `created`, `is_active`.
- **student_grade**: `id`, `user_id`, `course_id`, `grade`, `percent`.
- **discussion_thread**: `id`, `course_id`, `title`, `author_id`, `created`.
- **discussion_post**: `id`, `thread_id`, `body` (sanitised), `author_id`, `created`.
- **certificate_generated**: `id`, `user_id`, `course_id`, `created`, `download_url` (signed URL).
- **team**: `id`, `course_id`, `name`, `description`, `topic`.
- **team_membership**: `id`, `team_id`, `user_id`, `role`.
- **bookmark**: `id`, `user_id`, `course_id`, `content_id`, `created`.

0.2.3 Data Protection Measures

- Sensitive fields (email, password hash, personal data) are encrypted at rest using AES-256.
- Database backups are encrypted and stored separately.
- Audit logs are maintained for all critical database operations (user creation, role changes, grade updates).

0.3 Design Traceability

The following table maps requirements to API endpoints and database tables, with security annotations.

Requirement ID	API Endpoint(s)	Table(s)
UFR-OE-001-01	POST /api/courses/v1/courses	course_overview
UFR-OE-001-03	POST /api/enrollment/v1/enrollment	course_enrollment
UFR-OE-003-01	GET /api/discussion/v1/threads	discussion_thread
UFR-OE-003-02	POST /api/email/v1/bulk	(Celery)
UFR-OE-004-01	PUT /api/courses/v1/courses/{course_id}/content	course_structure
UFR-OE-005-03	POST /api/ora/v1/assessments	ora_assessment
UFR-OE-006-03	GET /api/gradebook/v1/grades	student_grade
UFR-OE-103-01	POST /api/enrollment/v1/enrollment	course_enrollment
UFR-OE-107-01	GET /api/certificates/v1/status	certificate_generated
UFR-OE-109-01	POST /api/discussion/v1/posts	discussion_post
UFR-OE-113-01	POST /api/bookmarks/v1/	bookmark
SEC-NFR-001..007	All endpoints	All tables (with security controls)

0.4 Activity Diagrams

0.4.1 Activity: Course Publishing

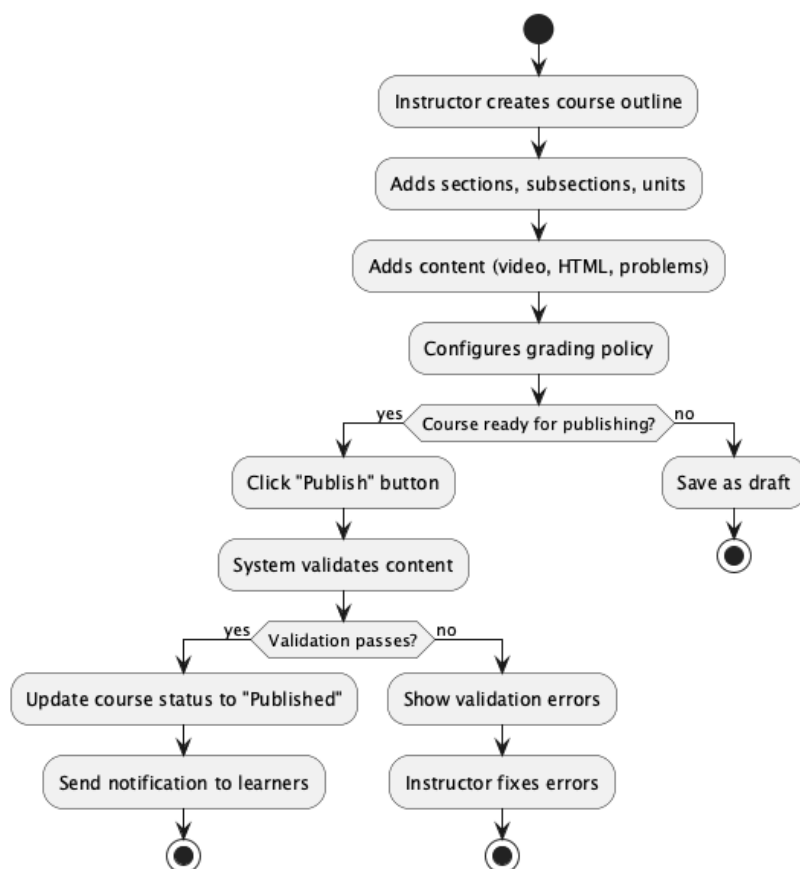


Figure 2: Course publishing workflow.

0.4.2 Activity: Open Response Assessment

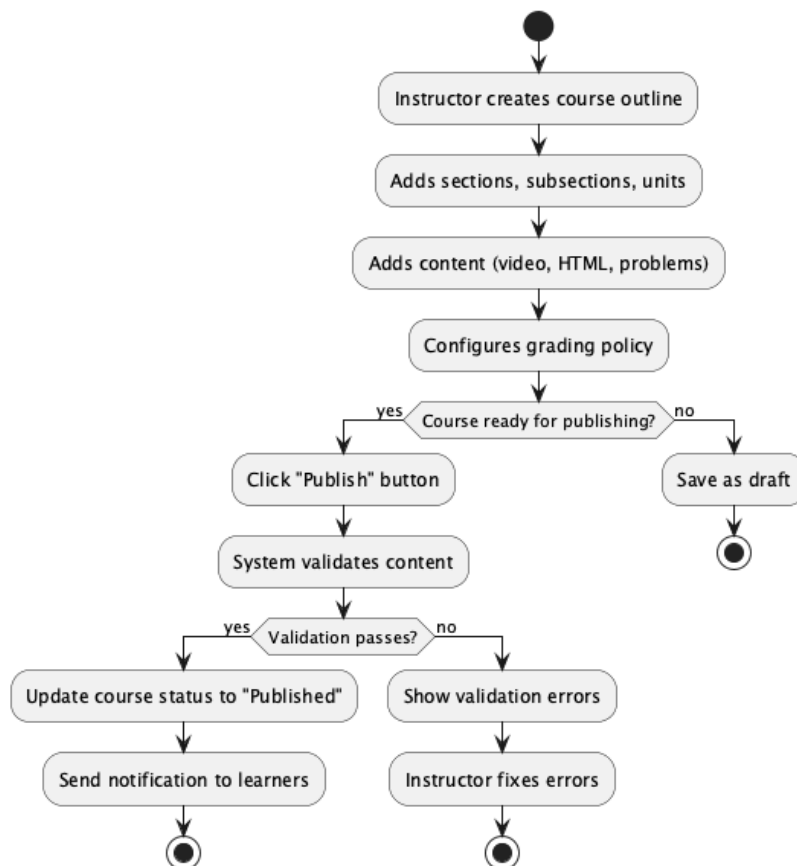


Figure 3: Open Response Assessment workflow.

0.5 References

- *Security Architecture & Threat Model – Moove Education Platform* (Security_OpenEdX.tex)
- CMCSRS v2.0
- Swagger API documentation (swagger-docs.json)

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